Predictive Modeling and Risk Scoring for Bank Customer Churn

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Abstract— Customer churn is a major challenge for retail banking because the loss of an existing customer can reduce recurring revenue and increase acquisition costs. This paper presents an end-to-end machine learning system for predicting bank customer churn and assigning interpretable risk scores. The project treats churn as a supervised binary-classification problem with *Exited* as the target, where 1 denotes churn and 0 denotes retention. The implemented workflow covers data validation, removal of identifier variables, categorical encoding, numerical preprocessing, business-oriented feature engineering, stratified train-test splitting, comparative model development, evaluation, explainability, model persistence, and interactive Streamlit deployment. The dataset contains 10,000 customer records and exhibits moderate class imbalance, with 20.37% churned customers and 79.63% retained customers. The final application is designed to return both a binary churn decision and a churn probability, enabling customer-level risk prioritization. The architecture emphasizes reproducibility, modularity, defensive validation, and consistent preprocessing between training and inference. The resulting system provides a practical foundation for targeted retention campaigns and data-driven customer relationship management.

Index Terms— customer churn, banking analytics, machine learning, binary classification, risk scoring, feature engineering, explainable AI, Streamlit

I. INTRODUCTION

Customer retention is an important business objective in retail banking. Customers interact with banks through multiple products and channels, and their likelihood of leaving may depend on financial characteristics, engagement, product usage, age, tenure, geography, and related behavioral patterns. A predictive system can help a bank move from reactive retention to proactive intervention by identifying customers with elevated churn probability before the relationship is lost.

The objective of this project is to build a reusable and production-oriented churn prediction workflow rather than a single experimental model. The system is organized into separate modules for preprocessing, exploratory analysis, feature engineering, model training, evaluation, explainability, visualization, persistence, and inference. The same saved preprocessing artifacts are reused during prediction so that the data transformation applied at inference is consistent with the transformation used during training.

The project documentation specifies supervised binary classification with *Exited* as the target. CustomerId and Surname are treated as identifiers rather than predictive variables. The prediction interface produces a churn probability and a threshold-based binary decision, while risk categories are derived from the predicted probability. This supports both technical evaluation and business-oriented prioritization.

The main contributions of the project are: (1) a modular end-to-end churn analytics pipeline; (2) business-relevant feature engineering with explicit leakage controls; (3) comparative evaluation of multiple classical machine learning models; (4) reusable model and preprocessing persistence; (5) customer-level probability and risk scoring; and (6) an interactive Streamlit interface for practical use.

II. PROBLEM FORMULATION

Let each customer be represented by a feature vector *x* containing demographic, financial, and behavioral attributes. The task is to learn a classifier f(x) that estimates the probability of churn and assigns a class label:

■ = f(x),   p = P(Exited = 1 | x)

where ■ is the predicted churn class and *p* is the estimated churn probability. A probability threshold is then used to convert the continuous risk estimate into a binary business decision. The project also maps probability ranges to Low, Medium, and High risk categories for easier operational interpretation.

The business objective is not simply to maximize accuracy. Because the dataset is moderately imbalanced, a model that predicts most customers as retained can still obtain high accuracy while failing to identify many true churners. Consequently, precision, recall, F1-score, ROC-AUC, and the confusion matrix are considered alongside accuracy. Recall is particularly important because missed churners may represent preventable customer loss.

III. DATASET AND EXPLORATORY ANALYSIS

The validated dataset contains 10,000 customer-level records. The target variable *Exited* contains 7,963 retained customers (79.63%) and 2,037 churned customers (20.37%). This distribution confirms moderate class imbalance and motivates the use of stratified sampling and multiple evaluation metrics.

Variable	Role / Type
CreditScore	Numerical
Geography	Categorical
Gender	Categorical
Age	Numerical
Tenure	Numerical
Balance	Numerical
NumOfProducts	Numerical / ordinal
HasCrCard	Binary
IsActiveMember	Binary
EstimatedSalary	Numerical
Exited	Target (0/1)

TABLE I. PRIMARY MODELING VARIABLES

Descriptive analysis reports a median credit score of 652 with values ranging from 350 to 850. Median customer age is 37 years, median tenure is 5 years, and most customers own one or two products. Median balance is approximately 97,199 and median estimated salary is approximately 100,194. The customer base spans France, Germany, and Spain, with France representing roughly half of the observations. Gender is reasonably balanced, with 5,457 male and 4,543 female customers.

Data-quality analysis found no missing values and no duplicate records in the documented dataset analysis. CustomerId and Surname are excluded from predictive modeling because they are identifiers; Surname also has high cardinality. Geography and Gender require categorical encoding, while numerical variables can be scaled when required by the selected algorithm.

Class	Count	Percentage
Retained (0)	7,963	79.63%
Churned (1)	2,037	20.37%

TABLE II. TARGET CLASS DISTRIBUTION

IV. DATA PREPROCESSING AND FEATURE ENGINEERING

The preprocessing stage is designed to make the raw customer table suitable for supervised learning while preserving a deterministic transformation path. Identifier fields are removed, categorical variables are encoded, and numerical variables are scaled where appropriate. The train-test strategy uses an 80/20 split with a fixed random state and stratification on *Exited*, preserving the churn distribution in both subsets.

Feature engineering is guided by business relevance, inference-time availability, interpretability, and leakage prevention. The documented feature candidates include BalancePerProduct, SalaryToBalanceRatio, CreditScoreCategory, AgeGroup, ProductsPerTenure, ZeroBalanceFlag, ActiveCreditCardCustomer, EngagementScore, CustomerLoyaltyScore, and FinancialActivityScore. Features that were redundant, subjective, or dependent on unavailable information were rejected. The leakage assessment confirms that approved engineered features use predictor variables available at inference time and do not reference the target or future outcomes.

The central design principle is to perform the same transformations during training and inference. The preprocessing pipeline is persisted with joblib and later loaded by the prediction module. This reduces training-serving skew and allows the Streamlit application to use the exact transformation logic validated during model development.

V. MACHINE LEARNING METHODOLOGY

The modeling stage compares several classical classification algorithms under the same train-test conditions. Logistic Regression is used as a baseline because of its simplicity and interpretability. Decision Tree and Random Forest models provide nonlinear decision boundaries and naturally capture interactions. Gradient Boosting is included as an additional ensemble approach. XGBoost is documented as optional when time and dependency constraints permit.

Model comparison is based on Accuracy, Precision, Recall, F1-score, and ROC-AUC. Accuracy measures overall correctness, Precision measures the fraction of predicted churners who actually churn, Recall measures the fraction of true churners identified, and F1 balances precision and recall. ROC-AUC evaluates discrimination across thresholds. The confusion matrix provides a direct view of true positives, true negatives, false positives, and false negatives.

The project documentation recommends Stratified K-Fold validation as an extension after baseline evaluation. Hyperparameter optimization is treated as a controlled step after establishing baseline performance, using grid or randomized search where beneficial. This avoids spending computational effort on tuning before a reliable baseline has been established.

VI. MODEL EXPLAINABILITY AND RISK SCORING

A banking prediction system should provide more than a class label. The project therefore includes feature-importance analysis and SHAP-based explainability as part of the planned model interpretation layer. Explainability is intended to help stakeholders understand the factors associated with a high-risk prediction and to support responsible use of the model.

The inference workflow loads the persisted model, preprocessing pipeline, and feature definitions; validates the customer input; applies the saved transformations; generates a class prediction; and, when supported by the model, calculates a probability and confidence score. The documented risk bands classify probability below 0.30 as Low, 0.30–0.70 as Medium, and above 0.70 as High. This converts a numerical prediction into an actionable prioritization scheme.

For batch inference, the system appends Prediction, Probability, Confidence, and Risk Level to the validated input data and can export the results as CSV. A prediction summary reports total customers, predicted churn, predicted retention, average probability, and counts of customers in each risk category.

VII. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The project follows a modular software architecture. The data-preprocessing layer loads and validates the source data. Exploratory analysis produces descriptive statistics and visualizations. Feature engineering prepares the modeling matrix and reusable preprocessing artifacts. Model training compares candidate algorithms. Model evaluation calculates standardized metrics. Explainability and visualization modules produce interpretable outputs. Model persistence stores the trained model, preprocessing pipeline, feature columns, metrics, and metadata. The prediction module provides single-record and batch inference.

Layer	Main Responsibility
Data	Loading, validation, cleaning
EDA	Distributions, correlations, insights
Features	Encoding, scaling, engineered variables
Training	Model fitting and comparison
Evaluation	Metrics, confusion matrix, ROC-AUC
Explainability	Feature importance and SHAP
Persistence	Model and preprocessing artifacts
Inference	Single/batch prediction and risk
UI	Streamlit dashboard and interaction

TABLE III. SYSTEM MODULES

The application layer is implemented using Streamlit. The planned dashboard includes an executive view, dataset/EDA analysis, model performance, customer churn prediction, feature importance, and an About Project page. The interface is intentionally separated from training logic so that deployment uses persisted artifacts rather than retraining models during application startup.

The project also follows production-oriented software practices documented across the modules: deterministic execution, centralized configuration, explicit validation, logging, type annotations, reusable utilities, exception chaining, and single-responsibility design. The persistence layer uses joblib for serialized artifacts and JSON for metrics and metadata.

VIII. RESULTS AND DISCUSSION

The documented empirical dataset analysis establishes several important characteristics of the problem. The 20.37% churn rate indicates that churn is the minority class, making class-sensitive evaluation essential. The feature distributions show meaningful variation in credit score, age, tenure, balance, product count, and salary, while the categorical variables provide geographic and demographic context.

The project implementation successfully establishes an executable inference pipeline. During integration, artifact-loading and input-validation issues were resolved, including model metadata persistence, model loading compatibility, missing Year input handling, and categorical risk-level assignment. The final prediction workflow is designed to run on both single customer records and batches, producing a probability and interpretable risk level.

Because the project source material available for this paper does not contain a final numeric model-comparison table from the completed training run, this paper deliberately does not invent accuracy, precision, recall, F1, or ROC-AUC values. The evaluation framework is implemented to generate these metrics from the final persisted model and test set. This distinction preserves the integrity and reproducibility of the reported results.

IX. BUSINESS IMPLICATIONS

The system can support a retention workflow in which customers are prioritized by predicted churn probability. High-risk customers can be reviewed first, followed by medium-risk customers for proactive engagement. Business teams can combine the risk score with customer context such as age, tenure, product count, activity, balance, and geography when designing retention actions.

A key benefit is resource allocation. Instead of contacting every customer with the same intervention, the bank can focus retention efforts on segments with elevated predicted risk. Feature-importance and SHAP explanations can further support analyst review by showing which input characteristics contribute to an individual prediction.

The model should be treated as decision support rather than an autonomous decision-maker. Probability thresholds should be calibrated against the bank's intervention costs, customer experience objectives, and acceptable false-positive/false-negative trade-offs. Monitoring should also be used after deployment to detect drift and changes in customer behavior.

X. LIMITATIONS

The project is based on a structured customer-level dataset and therefore inherits the limitations of the available variables. Historical customer behavior outside the recorded attributes is not directly represented. The model may also be affected by class imbalance, distribution shift, and changes in banking products or customer behavior over time.

The current project is intentionally scoped as an internship-level production-oriented ML application. REST API services, authentication, database integration, cloud deployment, automated retraining, distributed computing, and full MLOps orchestration are outside the implemented scope. These components are suitable future extensions rather than requirements for the current system.

XI. FUTURE SCOPE

Future work can extend the system in five directions. First, threshold optimization and probability calibration can improve the alignment between model scores and business intervention costs. Second, scheduled retraining and data-drift monitoring can keep the model current. Third, a REST API can expose the inference layer to external applications. Fourth, cloud deployment and containerization can improve scalability. Finally, richer customer interaction data can support temporal modeling and more advanced segmentation.

XII. CONCLUSION

This paper presents a modular predictive modeling and risk-scoring system for bank customer churn. The project frames churn as a supervised binary-classification problem, validates and preprocesses customer data, engineers business-relevant features, compares multiple machine learning algorithms, evaluates performance with class-sensitive metrics, persists reusable artifacts, and exposes customer-level prediction through Streamlit. The documented dataset contains 10,000 customers with a 20.37% churn rate, demonstrating why recall, F1, and ROC-AUC are important complements to accuracy.

The principal engineering outcome is a consistent path from raw customer information to deployable risk scores. By separating training, persistence, and inference responsibilities and by reusing the saved preprocessing pipeline, the project establishes a practical foundation for data-driven customer retention. The system can be extended with calibrated thresholds, monitoring, APIs, cloud deployment, and automated retraining while preserving the current modular architecture.

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